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A high-throughput pipeline for 3D reconstruction of soybean root system architecture using macro-CT scanning

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ABSTRACT. Root systems and their architectures are fundamental to plant development, resource acquisition, and stress resilience, yet root system phenotyping remains technically challenging because roots are hidden and the root systems architecture (RSA) is complex within opaque soil environments. In this study, we present a large-scale, non-destructive phenotyping framework for soybean root systems using a macro-CT scanner (CT: computed tomography) combined with a biologically informed 3D reconstruction pipeline. Soybean plants were cultivated in sand, a medium chosen for its favorable density contrast with root tissues and its compatibility with imaging-based root system analysis. An automated preprocessing pipeline first converted CT slice data into initial 3D root system point clouds, removing distant noise while leaving substantial noise near the roots. Subsequently, semi-automated particle tracking was conducted, initiated from manually segmented primary roots and selected lateral root tips. A distance-based pruning strategy was then employed to eliminate trajectories deviating into noise, while a neighborhood retrieval approach recovered root geometry around retained trajectories. Tested across diverse soybean genotypes in controlled greenhouse conditions, this approach robustly reconstructed coherent and biologically realistic RSAs from noisy CT datasets. This framework enables high-throughput, in-situ root system characterization at a scale large enough to be of value for modern breeding programs and provides a strong basis for further automation through deep learning-based primary root segmentation and root tip detection.

Keywords. Root system architecture, soybeans, 3D reconstruction, non-destructive phenotyping, computed tomography

Introduction

Root systems play a crucial role in plant development, resource acquisition, and resilience to environmental stresses (Ramachandran et al., 2024). Despite their fundamental importance, roots have historically received less attention than

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aboveground traits in plant research. However, with the growing urgency to develop climate-resilient crops and achieve sustainable agricultural practices, understanding root system architecture (RSA) has become a major focus in plant science (Lynch, 2022). Accurate characterization of RSA is critical for improving traits such as drought tolerance, nutrient use efficiency, and overall crop productivity. Yet, root phenotyping remains an outstanding technical challenge due to the hidden and complex nature of roots within opaque soil environments (Atkinson et al., 2019).

Root phenotyping methods are generally classified into destructive and non-destructive approaches. Destructive methods, such as excavating root systems from soil followed by structure-from-motion (SfM) 3D reconstruction, have been widely adopted due to their low cost and proven effectiveness (Liu et al., 2021; Schonberger & Frahm, 2016). However, the physical extraction process inevitably alters or damages the native root architecture, limiting the ability to capture true in situ morphology.

Alternatively, non-destructive methods, such as hydroponic growth systems, provide valuable opportunities to observe root development without soil interference (Clark et al., 2011). However, roots grown in hydroponic environments often display morphological characteristics distinct from those developed in soil, reflecting differences in mechanical support, chemical composition, and microbial interactions. Additionally, hydroponic systems are typically less suited for the large-scale cultivation of major commercial crops, which can constrain their applicability in breeding programs. While advances in hydroponic design may eventually bridge these gaps, current systems often do not fully replicate the complex conditions of field soils, limiting the direct transferability of phenotyping insights to real-world agricultural settings (Lopez-Guerrero et al., 2022).

Advanced imaging technologies, particularly micro-computed tomography (Micro-CT), have demonstrated outstanding capabilities for non-destructive, high-resolution visualization of root systems embedded within growth media (Mairhofer et al., 2015). Micro-CT provides micron-scale resolution, allowing for the precise characterization of fine root structures, root hairs, and branching patterns without disrupting the soil matrix (Nakhforoosh et al., 2024). This makes Micro-CT a valuable tool for detailed investigations into early root development, tissue structure, and root-soil interactions at small scales. Despite these advantages, several factors constrain the broader application of Micro-CT in large-scale plant phenotyping. The high spatial resolution comes at the cost of limited field of view, which restricts scanning to small volumes—typically suited only for young seedlings or miniature root systems (Mairhofer et al., 2012). Capturing the complete architecture of mature or larger crop root systems is often impractical without complex stitching of multiple scans, which further increases processing time and technical complexity. Moreover, the scanning process itself is relatively slow, with individual scans often exceeding 30 minutes per sample, posing a significant bottleneck when hundreds or thousands of plants must be assessed across different developmental stages.

To bridge this gap, we present a large-scale, non-destructive phenotyping framework for soybean root systems using Macro-CT scanning, specifically designed for high-throughput breeding contexts. By leveraging the fast scanning capabilities of a clinical-grade Macro-CT system, capable of imaging multiple plants simultaneously within minutes, we overcome the major bottlenecks associated with Micro-CT. Plants were cultivated in sand, a medium that provides sufficient mechanical support for natural root growth while offering a favorable density contrast against plant tissues, enabling effective separation of roots from the substrate using thresholding. Beyond data acquisition, we developed a novel reconstruction pipeline that integrates automated preprocessing, particle-based root tracking, path pruning, and structure recovery. This method fully respects the biological growth patterns of soybean roots, ensuring that in situ morphology is preserved throughout the reconstruction process.

Thus, this study presents a novel application of Macro-CT imaging combined with a semi-automated reconstruction framework for large-scale, non-destructive phenotyping of soybean root systems within a breeding program. By integrating high-throughput scanning with biological structure-aware tracking and reconstruction, our approach enables in situ characterization of complex root architectures at a scale and speed compatible with modern crop improvement pipelines.

Materials and Methods

Data collection. To enable non-destructive 3D imaging of soybean root systems across developmental stages, we cultivated 150 soybean plants in a controlled greenhouse environment (Figure 1a), representing a diverse set of genotypes. Each plant was labeled with a unique identifier corresponding to its genotype. For imaging, we employed a Macro-CT scanner (Canon Aquilion Prime SP), capable of scanning up to five pots simultaneously on the CT couch (Figure 1b). Each scan required approximately 5 minutes, allowing efficient throughput for large-scale experiments.

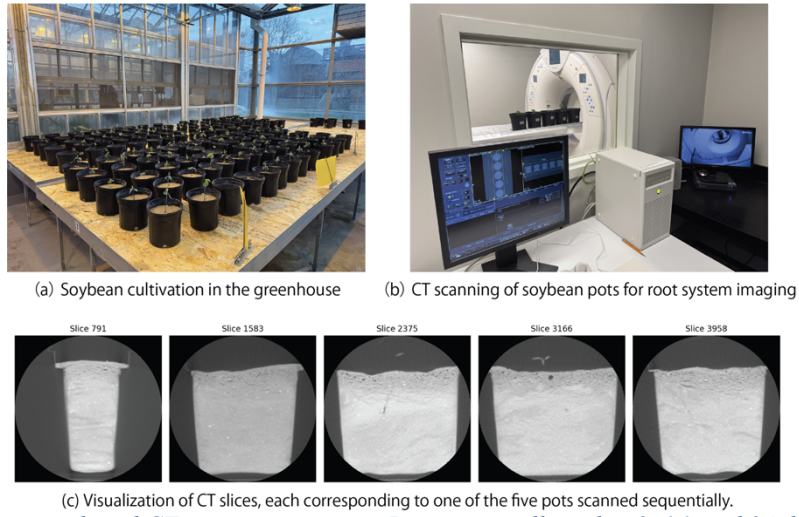


Figure 1: Soybean root growth and CT image acquisition. Data were collected at 8, 16, and 24 days after germination.

Compared to Micro-CT scanners, which typically require 30–60 minutes per scan depending on resolution and sample size, the Macro-CT approach offers a substantial advantage in scalability and speed, making it particularly suitable for high-throughput phenotyping in breeding programs. Each scan produced 4751 axial slices at a resolution of 512×512 pixels, with DICOM metadata indicating a pixel spacing of 0.390 mm in-plane and a slice thickness of 0.2 mm along the scanning axis (Figure 1c). These parameters enabled precise mapping from image coordinates to physical dimensions, yielding reconstructed volumes of $199.68 \text{ mm} \times 199.68 \text{ mm} \times 950.20 \text{ mm}$ (Figure 3b). Representative CT slices from a scan illustrate the diverse structure of plant and soil content across depths and pots, enabling downstream analysis of RSA.

Method overview. Figure 2 outlines the pipeline for reconstructing soybean root systems from CT-derived point clouds. The preprocessing stage is fully automated, including plant separation, artifact removal, density-based thresholding, and clustering-based denoising. While this step effectively removes noise far from the root systems, it does not fully clean the dense noise closely surrounding the roots. Cleaning these surrounding noise points remains highly challenging due to the similar density and spatial proximity to root structures.

To address this, we developed a particle tracking framework. After manually segmenting the primary root and selecting lateral root tips, tracking is initiated from each tip toward the primary root. A pruning step eliminates trajectories that deviate into noise, and a neighborhood retrieval process recovers the full lateral root morphology along the traced paths. This framework integrates automated preprocessing with biologically informed, semi-automated tracking to reconstruct high-fidelity RSAs from noisy CT data.

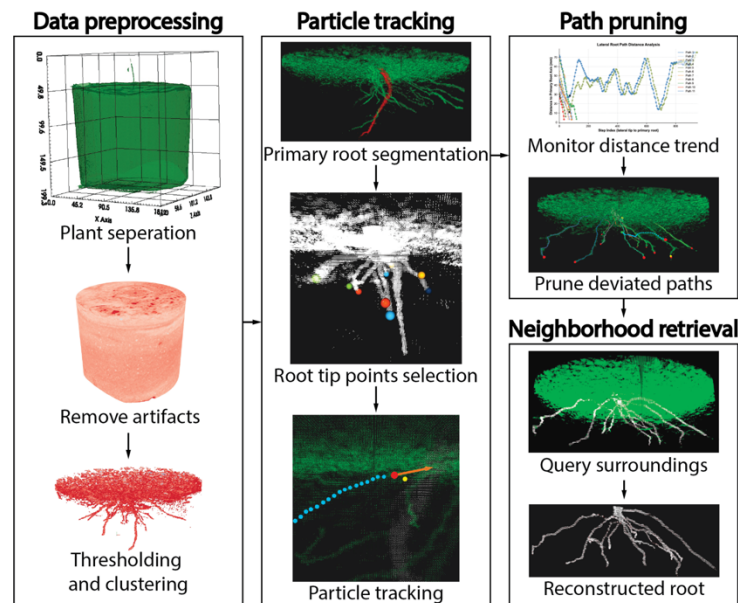
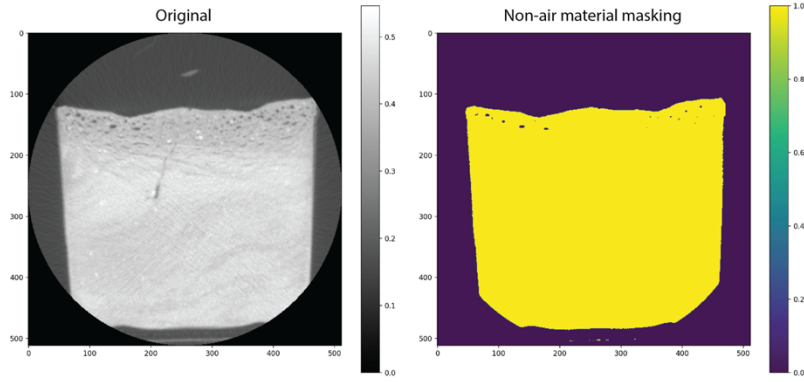
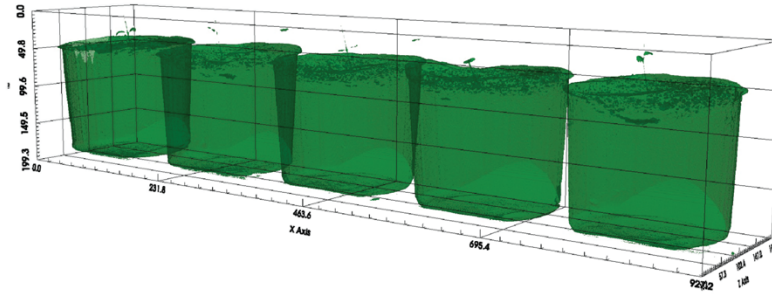


Figure 2: Overview of the root system reconstruction pipeline. The pipeline includes data preprocessing, particle tracking from root tips, pruning of deviated paths, and neighborhood retrieval to recover complete lateral root structures.

Non-air masking. To distinguish physical material from air in the CT data, we applied Otsu’s thresholding method slice-by-slice to the reconstructed volumetric dataset (Otsu, 1975). Otsu’s thresholding is a widely used, unsupervised method for automatic image binarization, particularly effective in segmenting images with bimodal histograms. This automated technique efficiently segments the air phase by identifying an intensity threshold that maximizes inter-class variance, producing a binary mask that delineates the pot and plant material (Figure 3a). Following segmentation, we reconstructed the 3D volume of the entire scan, which typically encompassed five soybean pots imaged simultaneously, though occasionally fewer were present due to approximately 10% plant mortality. To ensure accurate physical interpretation, the reconstruction incorporated spatial metadata from the DICOM headers, using a pixel spacing of 0.390 mm in the YZ plane and a fixed slice thickness of 0.2 mm along the X-axis. These parameters enabled the precise conversion of slice-based indices to physical coordinates in millimeters, allowing the resulting point cloud (Figure 3b) to accurately reflect the true spatial dimensions of the scanned specimens. This physically calibrated volume forms the foundation for subsequent phenotyping and geometric analysis of RSA.



(a) Otsu thresholding applied to segment air from material in CT slices

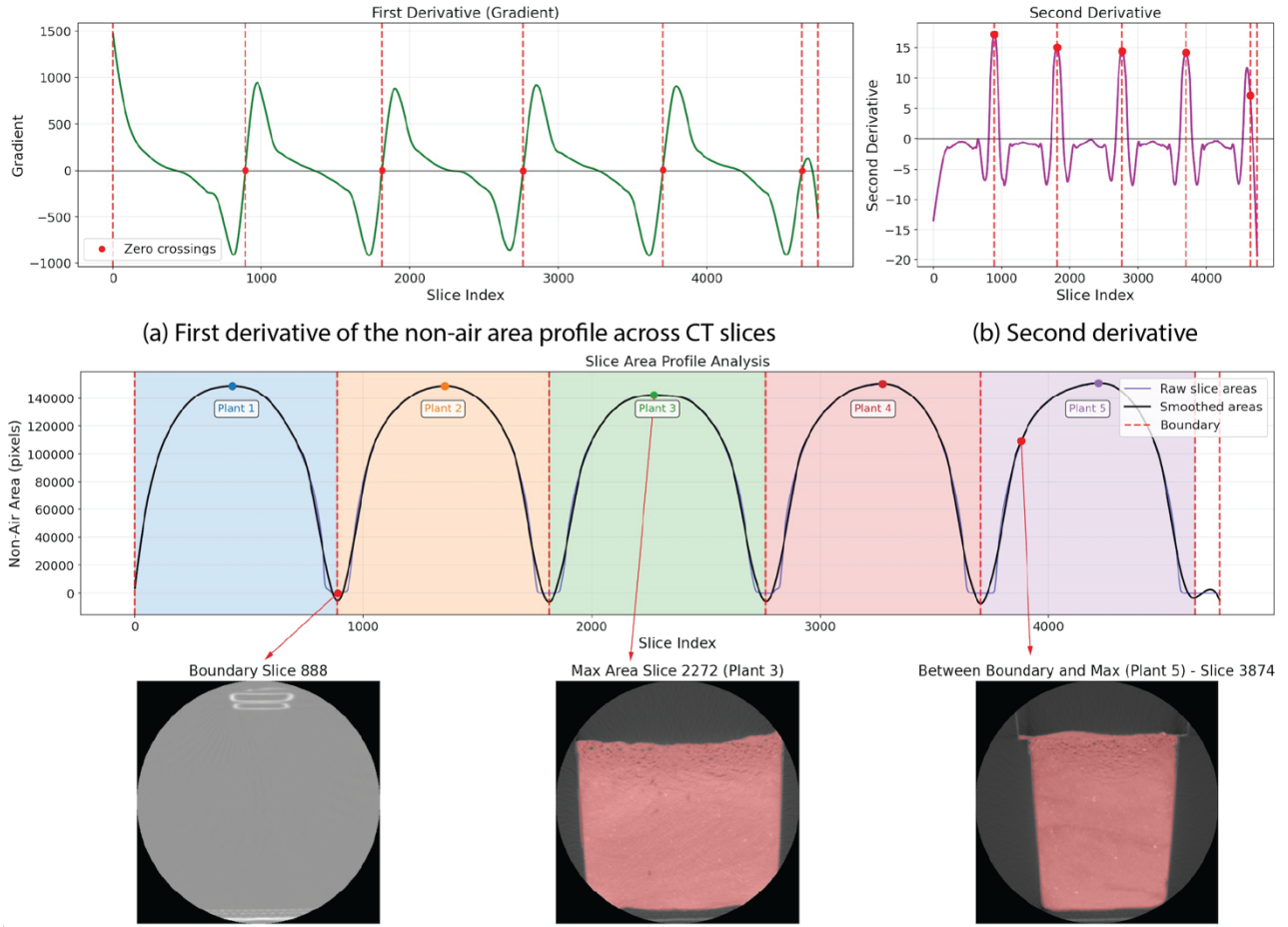


(b) Reconstructed 3D volume of soybean pots from a single CT scan, shown in actual physical dimensions

Figure 3: Non-air masking and 3D reconstruction of soybean pots.

Plant separations. To enable scalable and plant-specific root system analysis, it is essential to isolate individual soybean pots from batch-scanned CT volumes. Rather than applying downstream processing to entire sequential scans—which may contain multiple plants—our goal is to segment and analyze each plant separately. To achieve this in a fully automated manner, especially given the large sample size in high-throughput phenotyping experiments, we leverage the characteristic profile of non-air regions along the scanning axis. After applying Otsu’s thresholding to segment non-air material (roots, soil, and pots), we compute the area of non-air regions slice-by-slice. We observed a consistent spatial pattern in which each pot’s profile exhibits a gradual increase in non-air area, reaches a peak near the middle of the pot, and then diminishes before the next pot begins.

Figure 4 illustrates our method for exploiting this pattern. The first and second derivatives of the smoothed area profile (Figure 4a–b) are used to identify valley points—transitions between neighboring pots—based on zero crossings and positive curvature. These boundaries are then used to segment individual pots. As shown in Figure 4c, raw and smoothed area profiles highlight each plant’s cross-sectional footprint, and example CT slices demonstrate distinct boundary (left), center (middle), and transitional (right) regions. This automated pipeline robustly segments individual plants from large-scale scans, forming a critical preprocessing step for subsequent root system analysis.



(c) Non-air area profile across slices, showing raw (gray) and smoothed (black) curves.
Example CT slices highlight a boundary (left), peak area (center), and transitional region (right).
Figure 4: Slice area profile and derivative analysis used to identify individual soybean pots in CT scan data.

Artifacts removal. We separated each CT scan into individual plant point clouds before further processing. However, structural artifacts from the pot and surface artifacts from the air-sand intersection layer exhibited density ranges similar to those of the roots, making them difficult to distinguish through simple thresholding. Therefore, a dedicated preprocessing workflow was developed to remove these artifacts while preserving the integrity of the root system (Figure 5).

First, a hollow cylinder was fitted around the outer boundary of the point cloud to identify and exclude the pot structure (Figure 5a). The cylinder was parameterized based on the average radial distance in the X-Z plane, with adjustments for pot wall thickness and potential deviations from perfect circularity. Points falling within the pot wall boundaries, along with points located near the bottom of the pot, were removed from the dataset.

After pot removal, the point cloud primarily contained the sand medium and the embedded root structure (Figure 5b). To address surface artifacts caused by the air-sand interface, the vertical distribution of points was analyzed by dividing the cloud into horizontal layers and evaluating the point density within each layer. The topmost dense layer, corresponding to the air-sand intersection, was identified and removed along with a small margin below it to ensure clean separation (Figure 5c). The final cleaned point cloud retained only the sand-embedded root system without structural and surface artifacts (Figure 5d). This preprocessing step was essential to enable reliable root extraction in subsequent analyses.

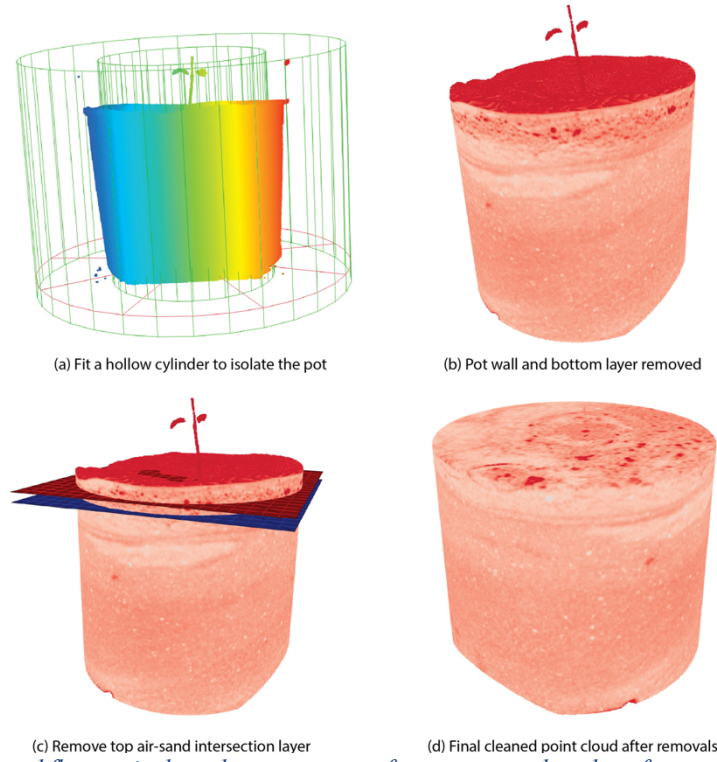
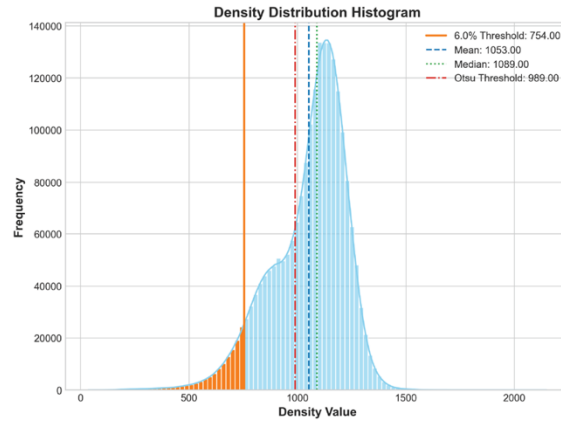
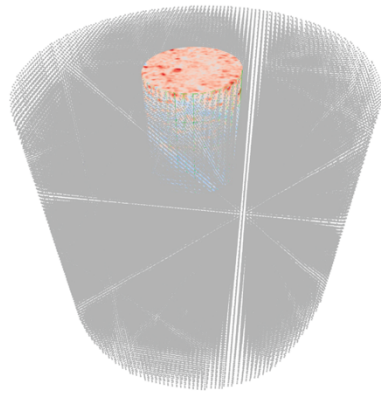


Figure 5: Preprocessing workflow to isolate the root system from structural and surface artifacts in CT-reconstructed point clouds.

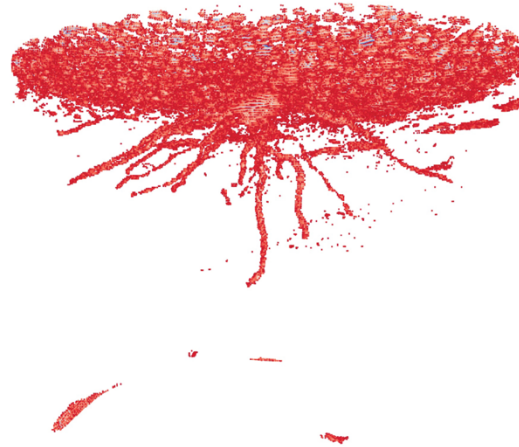
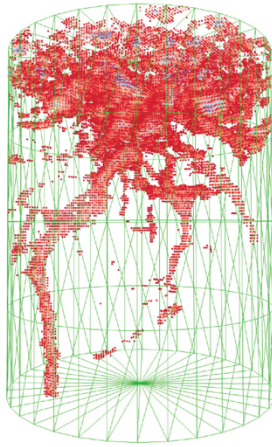
Root extraction via center-top cylinder density analysis. To extract the root system from the CT-reconstructed point clouds, we developed a density-based filtering strategy guided by the local density distribution (Figure 6). First, a cylindrical volume was selected from the top center of the point cloud (Figure 6a), targeting a region expected to contain predominantly root tissues and a minimal amount of surrounding medium. The cylinder diameter (40 mm) and height (50 mm) were predefined to ensure consistent sampling across different scans.

The density distribution within the selected cylinder was then analyzed to determine an appropriate threshold for root extraction (Figure 6b). A histogram of density values was constructed, and multiple statistical criteria were evaluated, including the mean, median, Otsu threshold, and a fixed percentile threshold set at 6%. Although Otsu’s method is a widely used approach for thresholding, it assumes the presence of distinct classes separated by clear peaks in the histogram. However, the density distributions observed in our data typically exhibited an unbalanced unimodal shape without distinct peaks, making it difficult for Otsu’s method to accurately determine a separation threshold. Consequently, we empirically selected the 6% percentile threshold, which reliably captured the lower-density signature of root tissues while minimizing the inclusion of surrounding noise and sand particles. Selecting the lowest 6% of densities effectively preserved fine root structures without aggressive loss or over-inclusion of background points, providing a balance between specificity for root identification and general applicability across samples.

Using the determined threshold, points within the cylinder were filtered to visualize the initial root extraction (Figure 6c). Finally, the threshold was applied to the entire pot-removed point cloud to extract the full root system from the remaining volume (Figure 6d). This cylinder-informed thresholding method improved robustness across samples with varying density distributions and minimized manual intervention.



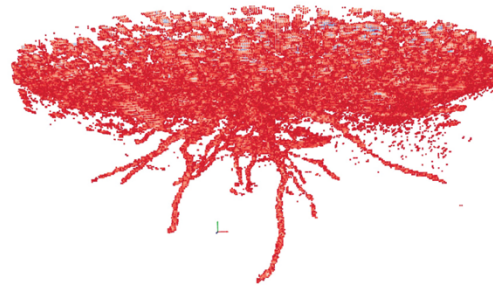
(a) Select a cylinder region from the top center (b) Analyze cylinder density distribution to determine root threshold



(c) Density-based filtering within the cylinder (d) Apply the analyzed threshold to the pot-removed point cloud

Figure 6: Root extraction using density thresholding informed by a top-center cylindrical region.

Root system refinement via clustering-based noise removal. Following density-based extraction, discrete noise artifacts still remained surrounding the root system (Figure 7a). To further refine the segmentation, we applied HDBSCAN clustering to the filtered point cloud. HDBSCAN is a hierarchical density-based clustering algorithm that is robust to varying point densities and can effectively identify sparse noise points that do not belong to the coherent root structure (Malzer & Baum, 2020).



(a) Remove discrete noise by clustering

(b) Extracted root system

Figure 7: Clustering-based noise removal to refine root system reconstruction.

Clusters were assigned based on local point densities, with small or isolated clusters corresponding to noise, and the main root system forming the largest continuous cluster. To automate root system identification, we developed a scoring strategy that considered both the number of points and their spatial distribution relative to a predefined reference cylinder (matching the center-top extraction region). Specifically, for each cluster, we calculated the number of points falling inside the reference

cylinder and weighted this number by the overall cluster size to favor large, spatially coherent clusters anchored near the center. The cluster achieving the highest score was automatically selected as the root system. This approach effectively avoided mistakenly selecting large off-center noise aggregations, especially in complex scans with scattered debris.

After clustering, only the identified main root cluster was retained (Figure 7b), while smaller or spatially inconsistent noise clusters were discarded. This clustering-based processing substantially improved the purity, structural coherence, and biological relevance of the extracted root systems, minimizing manual intervention and ensuring consistency across different samples. However, clustering alone was insufficient to fully clean the data, as noise points that were closely adjacent to or attached to root structures remained difficult to remove. These residual noise points required further refinement through biologically informed tracking and reconstruction strategies.

Lateral root path reconstruction via adaptive particle tracking. Accurately reconstructing individual lateral root trajectories from noisy point clouds poses significant challenges, particularly due to wetness caused by irrigation—as water exhibits a similar density range to plant tissues—and the lower compactness of the top surface layers, where reduced sand density generates massive noise. To address these issues, we developed a particle tracking-based method that adaptively follows the local geometry of each lateral root from its tip to its junction with the primary root (Figure 8).

Prior to lateral root tracking, the primary root was manually segmented from the full root system point cloud using software CloudCompare (Girardeau-Montaut, 2024). This manual segmentation ensures a clean and reliable reference for determining when a lateral root trajectory terminates upon reaching the primary root surface (visualized as white clusters in Figure 8).

After selecting a lateral root tip (visualized as a red sphere in Figure 8a), the tracking process is initialized by estimating the local growth direction. This is achieved by extracting a local neighborhood around the selected tip and performing principal component analysis (PCA). The first principal component, representing the principal elongation axis of the local structure, provides the initial tracking direction. To ensure the tracking proceeds toward the primary root rather than away from it, the direction is validated by examining the local neighborhood density along both forward and backward directions.

Tracking then proceeds iteratively. As shown in Figure 8b, from the current tracked point (red sphere), a new point is predicted by moving a fixed step size (1 mm) along the current direction vector. The predicted point, without calibration, is indicated by a yellow sphere. However, small deviations are inevitably introduced during each step due to minor inaccuracies in the local geometry estimation and prediction process itself. Without correction, these small errors would accumulate over multiple steps, leading the tracking path to drift progressively away from the true lateral root trajectory.

To mitigate this accumulation of tracking error, a calibration step is incorporated at each iteration (Figure 8c). After prediction, a local neighborhood is sampled around the tracked yellow point, and the next position is corrected by moving to the density center of the neighborhood, shown as the green sphere. This correction is essential to maintain stable and accurate path progression, ensuring the tracker remains aligned with the physical root structure even in the absence of severe external noise.

Furthermore, at each step, the tracking direction is dynamically updated to adapt to natural curvatures of the lateral root. After each calibrated move, a new local PCA is performed on the updated neighborhood to re-estimate the primary growth direction. To maintain smooth directional transitions and avoid abrupt changes caused by local point cloud irregularities, the new principal direction is blended with the previous movement vector using a smoothing factor. This combination of real-time direction adaptation and calibration ensures the tracker can robustly follow intricate and curved root paths through the 3D point cloud.

As tracking progresses (Figure 8d), the reconstructed path grows continuously, ultimately terminating when the traced trajectory approaches the manually segmented primary root within a predefined proximity threshold (Figure 8e). The full reconstructed lateral root trajectory is displayed in Figure 8f, starting from the manually selected tip and following a biologically plausible path back to the primary root, despite the complexity and local noise present in the dataset.

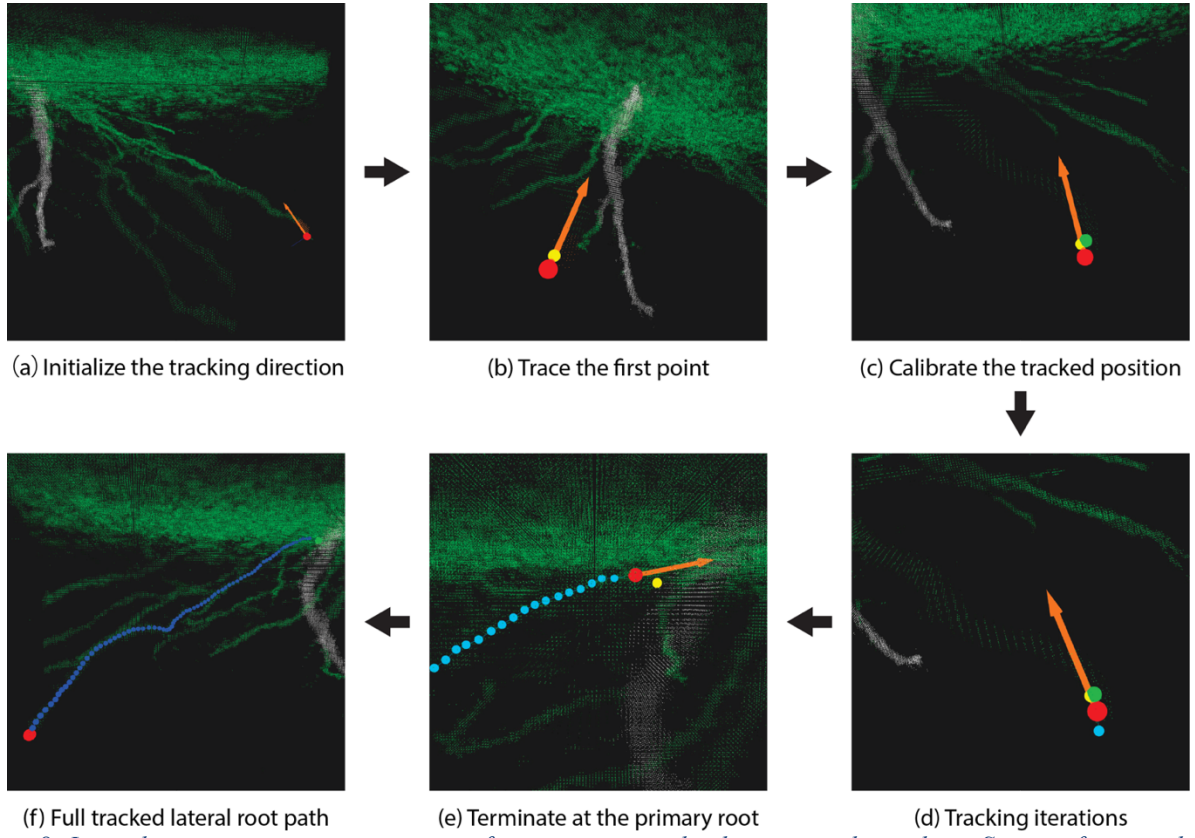


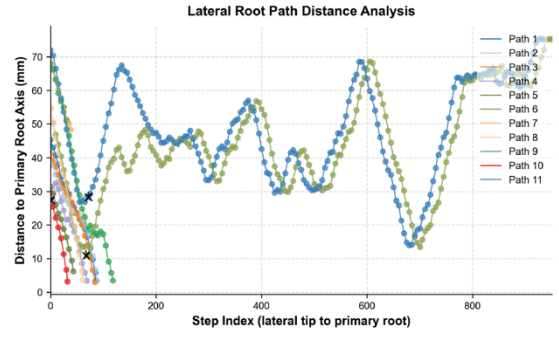
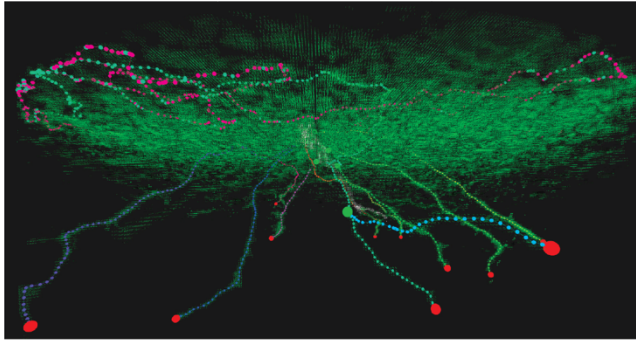
Figure 8: Lateral root trajectory reconstruction from noisy point clouds via particle tracking. Starting from a selected root tip, the method estimates direction, tracks and calibrates points, and terminates at the primary root.

Path pruning. Tracking lateral root trajectories from the root tip toward the primary root is generally reliable but can occasionally encounter errors, particularly when dense noise or voids are present. To enhance robustness against these rare failure cases, we developed strategies to monitor trajectory behavior and prune deviations. Observations revealed that erroneous trackings tend to violate a key biological principle: as tracking progresses, each step should move closer to the primary-lateral root junction. Ideally, the distance between the tracked point and the primary root should gradually decrease throughout the trajectory. In contrast, if tracking deviates sideways into noise, this distance will progressively increase. Therefore, we designed a method to continuously monitor distance trends during tracking and detect deviations based on sustained increases.

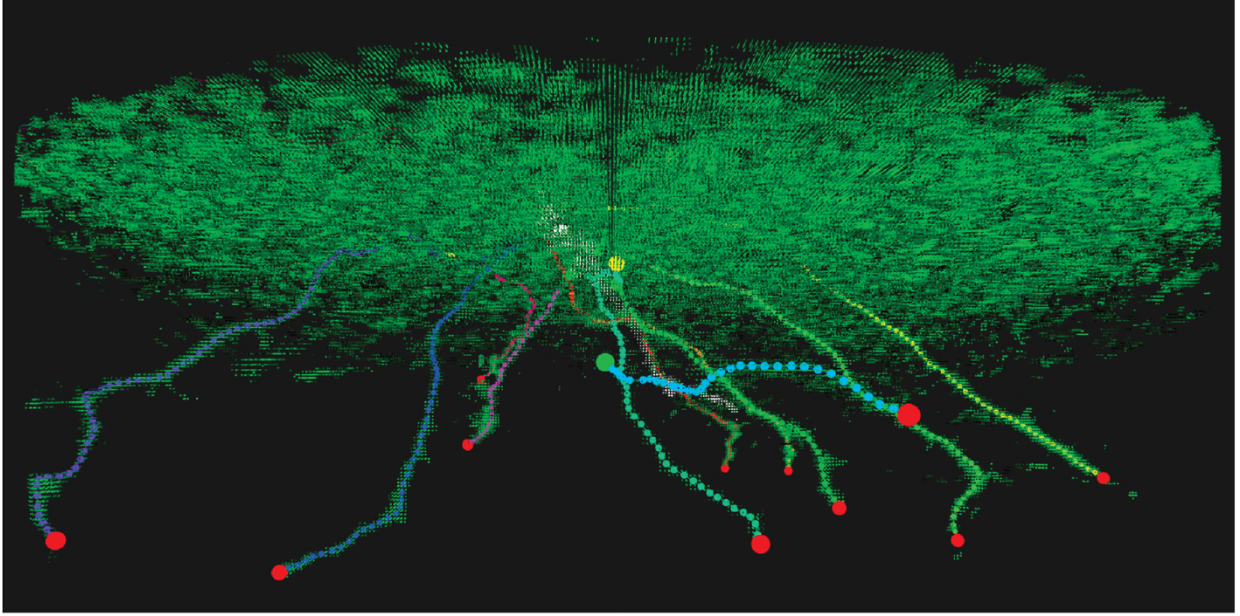
However, we are lack of prior knowledge of exact position of the primary-lateral root joint, we need to find a reliable method to estimate the distance between each tracked point and primary-lateral joint. To approximate the target position, we first extract the primary root axis using PCA, where the first principal component represents the direction of maximum variance among primary root points. Each tracked point is projected onto this axis to identify its closest location along the main root structure. The orthogonal distance—defined as the shortest distance from the tracked point to the axis—is then calculated as the Euclidean norm of the vector difference between the point and its projection. This distance quantifies how far a tracked point deviates from the ideal path toward the primary root. Although this projection does not recover the true joint location, it provides a practical proxy for assessing trajectory accuracy. If tracking remains correct, the distance to the primary axis should generally decrease or fluctuate slightly; in contrast, persistent increases in distance often indicate that the tracker has deviated into noise or voids.

Based on this observation, we introduce a pruning strategy that terminates a path if eight consecutive steps of distance increase are detected. The threshold of eight steps (equivalent to approximately 8 mm in our tracking resolution) provides a tolerance for the natural curvature of lateral roots, allowing paths to temporarily deviate without being prematurely pruned. This tolerance accommodates legitimate bending patterns while effectively filtering out extreme deviations likely caused by tracking drift into noise or voids.

Figure 9a illustrates the raw tracking results, where some paths deviate sideways into noise and fail to reach the primary root. The corresponding distance profiles are shown in Figure 9b, highlighting the trends of decreasing or increasing distance across steps. After applying the pruning strategy, as shown in Figure 9c, the lateral root trajectories more reliably converge toward the primary root axis, improving the overall fidelity of RSA reconstruction.



(a) Raw tracking results with paths deviating into noise and voids. (b) Lateral root distance to primary root axis plot.



(c) Final pruned paths converging toward the primary root.

Figure 9: Path pruning based on tracked point distance trend to the primary root axis to improve lateral root trajectory reconstruction.

Path neighborhood retrieval for root system reconstruction. To reconstruct the complete lateral root system structure from noisy CT-derived point clouds, we retrieved root points surrounding each pruned path. After pruning unreliable paths (as described in Figure 9), the retained paths serve as skeletonized trajectories through the true root system.

For each point along a pruned path, we collected neighboring points within a fixed search radius from the surrounding point cloud. This process leverages the spatial coherence of real root material around the path, whereas noise points are typically more scattered. By accumulating local neighborhoods along the entire trajectory, we were able to recover the lateral root geometry while minimizing noise interference.

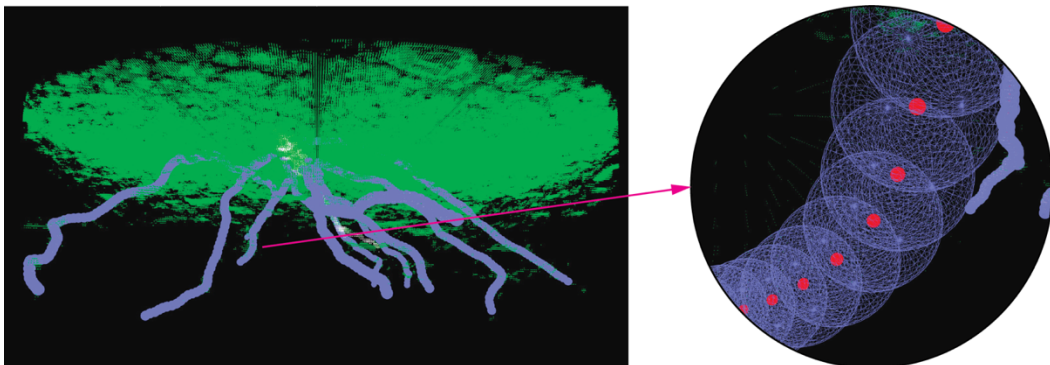


Figure 10: Root system reconstruction through neighborhood retrieval from noisy CT-derived point clouds.

Specifically, a 1.0 mm radius was applied for neighborhood retrieval, balancing the need to capture the full thickness of the roots while limiting the inclusion of extraneous noise. Figure 10 (left) shows the overall reconstructed root system after neighborhood retrieval, while the magnified view (right) highlights the sequential search regions (visualized as translucent spheres) around path nodes (red spheres). This recovery strategy enables robust reconstruction of continuous root branches even under noisy and fragmented CT imaging conditions.

Results and Discussion

We applied the proposed preprocessing, particle tracking, and reconstruction framework to CT-derived datasets of soybean root systems. Four representative examples are presented in Figure 11, illustrating the final reconstructed root system point clouds. Despite substantial noise remaining after threshold-based preprocessing, the integrated path tracing, pruning, and neighborhood retrieval methods enabled the recovery of coherent and biologically realistic root architectures. The reconstructed systems exhibited clear primary roots and well-separated lateral branches, faithfully capturing the overall spatial structures and growth patterns. Importantly, all components of our method were developed with full respect to the intrinsic morphology of soybean root systems, leveraging their biological growth tendencies—lateral roots emerging from and growing outward from the primary root axis. The effectiveness of our particle tracking approach, which follows the natural progression from lateral root tips toward the primary root, is critical to reliably reconstructing complex root systems. In particular, the pruning strategy successfully removed paths deviating into noise, while the neighborhood retrieval step restored full root thickness along each trajectory, suppressing scattered noise points.

The robustness of the developed framework is especially vital for large-scale experiments, where genotype differences and environmental variations—such as fluctuations in sunlight exposure time and angle—create uneven wetness within the growth medium. These variabilities introduce inconsistent noise levels during CT imaging, which are difficult to control or eliminate entirely. Given that such variability is unavoidable in practical experimental settings, a noise-resilient reconstruction approach is essential. Our results demonstrate that the proposed biologically-informed tracking and reconstruction framework can robustly recover complete and morphologically plausible RSAs even under highly challenging imaging conditions, providing a powerful tool for reliable root phenotyping at scale.

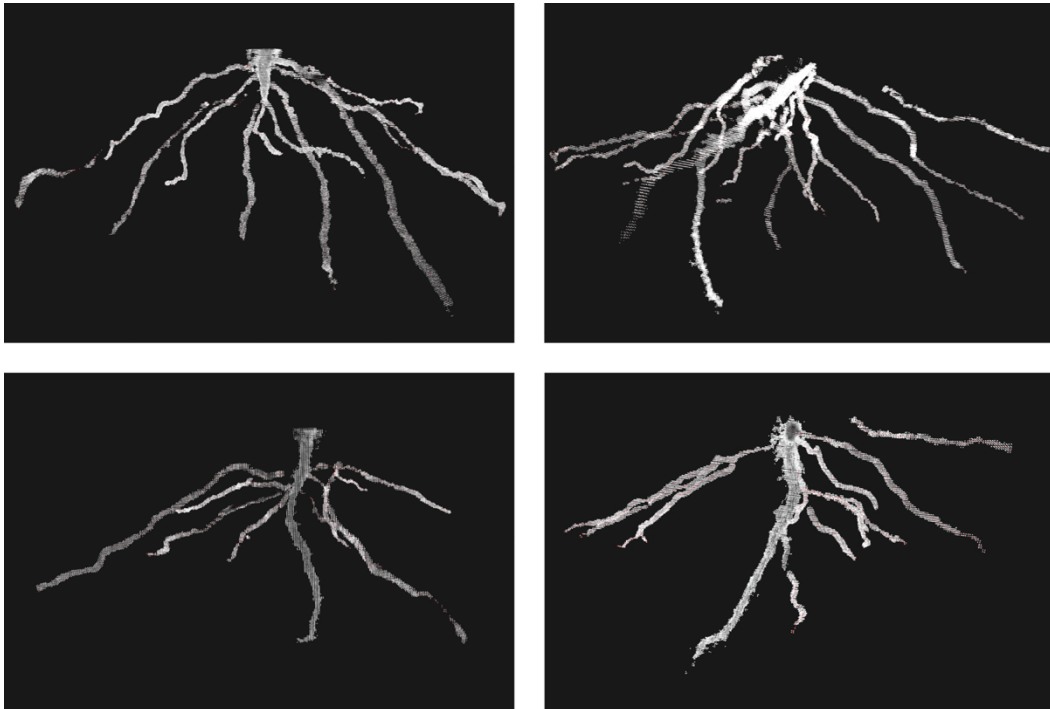


Figure 11: Four samples reconstructed into root system point clouds from CT data.

Limitation and future works. While the proposed framework successfully reconstructs complete and biologically plausible RSAs from noisy CT-derived point clouds, several limitations remain. Currently, the process is not fully automated: the root tip points for lateral roots must be manually selected, and the primary root needs to be manually segmented prior to tracking. To address this, we are developing a fully automated pipeline by training a root tip detection network and a primary root segmentation model, both based on Point Transformer architectures (Wu et al., 2024; Zhou et

al., 2024). These models will eliminate the need for manual intervention, allowing the entire reconstruction workflow to be automated and scalable. We have explored the alternative strategy of manually removing noise and labeling points into two classes—root versus non-root—but this approach proved to be highly labor-intensive and impractical for large datasets, and is therefore not recommended.

Another current limitation is the premature termination of some tracking paths. When trajectories deviate sideways into dense noise regions, our pruning strategy halts the tracking to prevent further drift, causing some paths to terminate before reaching the primary root. To further enhance root system completeness, we plan to bridge these gaps by leveraging curvature-based path reconstruction. Specifically, we aim to detect candidate endpoints and interpolate smooth paths based on local curvature continuity and directional consistency relative to the primary root axis. This approach would allow us to recover missed connections, ensuring a more complete RSA that will support more accurate downstream phenotyping analyses. Overall, these planned improvements will further increase the automation, robustness, and biological fidelity of our reconstruction framework.

Conclusions

In this study, we developed a novel Macro-CT-based framework for large-scale, non-destructive phenotyping of soybean root systems, addressing the longstanding challenges of imaging and reconstructing complex RSAs in-situ. By integrating high-throughput CT scanning with a biologically informed reconstruction pipeline—comprising automated preprocessing, particle tracking, path pruning, and neighborhood retrieval—we successfully recovered coherent and structurally faithful root architectures even under challenging imaging conditions. Our method fully respects the intrinsic growth patterns of soybean roots, enabling reliable reconstruction despite the presence of substantial noise and environmental variability.

Although the current framework requires manual selection of lateral root tips and primary root segmentation, ongoing efforts to train Point Transformer-based detection and segmentation models will enable full automation. Furthermore, future improvements will focus on bridging incomplete trajectories by leveraging curvature-based path interpolation to enhance root system completeness. Collectively, this work provides a scalable, robust, and biologically grounded solution for high-throughput RSA phenotyping, supporting the next generation of breeding programs aimed at developing climate-resilient crops.

Acknowledgments

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